

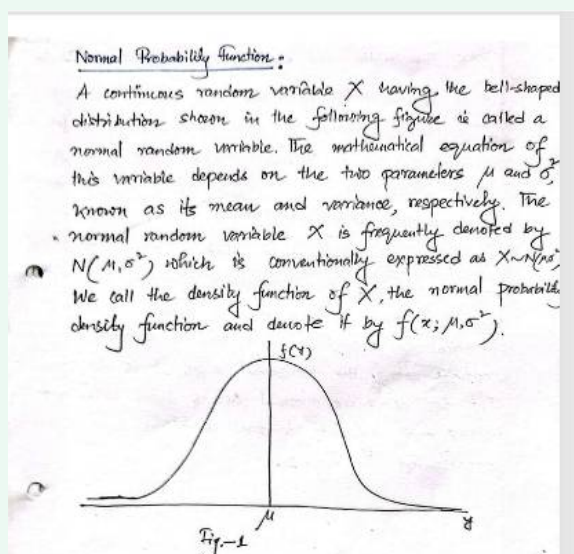


I will give you screenshots of some handwritten math notes. You will explain me that topic in the way it is in the note and give intuition for better understanding. OK?

Of course! 👍 Please upload the screenshots of your handwritten math notes, and I'll carefully read them. Then I'll:

1. **Explain the topic step by step** exactly as it appears in your notes.
2. **Add intuition and visualization** so you really understand *why* it works, not just *how*.
3. If needed, I can also make diagrams, solved examples, or flowcharts to make things clearer.

Go ahead and share the first screenshot.



Perfect 🧡 thanks for sharing the note. Let's go through it step by step in the same flow as your handwritten content, and I'll add intuition so it becomes crystal clear.

Topic: Normal Probability Function

From your notes:

1. Definition

- A continuous random variable X having the **bell-shaped distribution** (the curve in Fig-1) is called a **normal random variable**.
- Its equation depends on **two parameters**:
 - $\mu \rightarrow$ the **mean** (center of the curve)
 - $\sigma^2 \rightarrow$ the **variance** (spread or width of the curve)

2. Notation

- We write:

$$X \sim N(\mu, \sigma^2)$$

This means " X follows a normal distribution with mean μ and variance σ^2 ".

3. Density Function

- The probability density function (PDF) is written as:

$$f(x; \mu, \sigma)$$

which describes the shape of the bell curve.

4. Graph (Fig-1)

- The curve is **symmetric** around μ .
- The highest point (peak) is at $x = \mu$.
- As you go away from μ , the curve goes down smoothly but never touches the x-axis.

Intuition Behind It

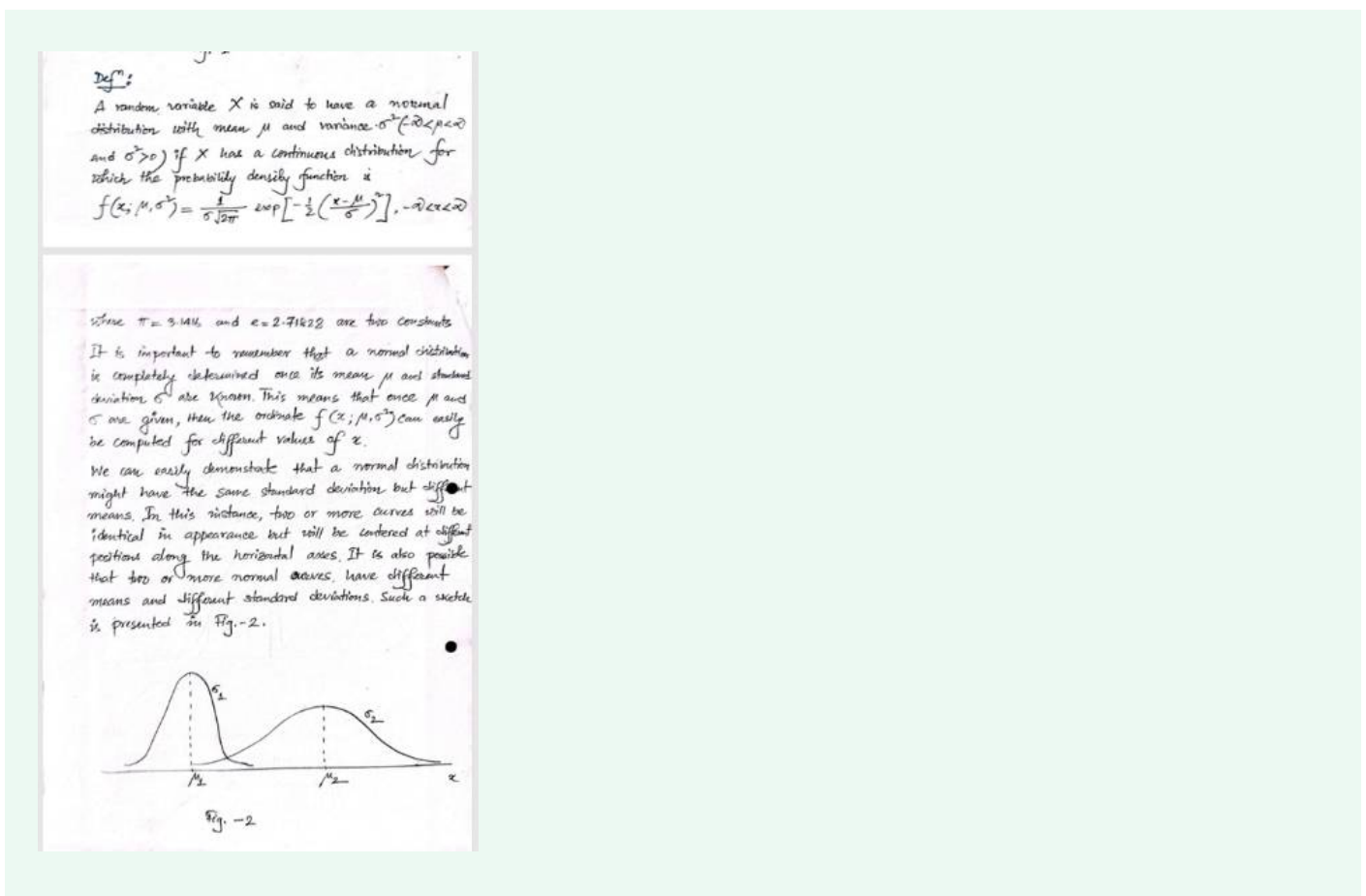
- Think of **heights of students in a class**. Most students are around the average height (say 165 cm). Very few are extremely short or extremely tall.
- That's why the curve is high in the middle (near the average) and falls off on both sides.
- **Mean (μ)**: Where the center of the data lies.
- **Variance (σ^2)**: How spread out the data is.
 - Small $\sigma \rightarrow$ narrow, tall curve (data tightly clustered).

- Large $\sigma \rightarrow$ wide, flat curve (data spread out).

✓ Key takeaway:

The normal distribution is one of the most important concepts in statistics because many real-life quantities (heights, exam scores, measurement errors) naturally follow this bell-shaped distribution.

👉 Do you want me to also **derive the exact formula** of the normal probability density function (the big exponential formula with $e^{-\frac{(x-\mu)^2}{2\sigma^2}}$) step by step, or should I stick to intuition and examples for now?



Great, now you've given the second part of your notes. Let's carefully go through it, explain the formula, and add intuition.

Definition (from your note)

A random variable X is said to have a **normal distribution** with mean μ and variance σ^2 (notation: $X \sim N(\mu, \sigma^2)$) if its **probability density function (PDF)** is:

$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right], \quad -\infty < x < \infty$$

Breaking the Formula Down

1. $\frac{1}{\sigma\sqrt{2\pi}}$

- This is a **scaling constant** to make sure the total probability under the curve is **1** (since probabilities must sum to 1).
- $\pi = 3.1416$, $e = 2.71828$.

2. $\exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$

- This part gives the **bell shape**.
- It decreases rapidly as x moves away from μ .
- The squaring makes the curve symmetric about the mean.

3. μ (**mean**)

- Decides where the center of the curve is (the "location").

4. σ (**standard deviation**)

- Decides how wide or narrow the curve is (the "spread").

Important Point in the Notes

- A normal distribution is **completely determined** by its mean μ and standard deviation σ .
 - Once you know these two, the whole curve is fixed.
- If two normal curves have:
 - **Same σ , different μ** : Curves look identical but shifted left or right.
 - **Same μ , different σ** : Curves centered at the same place but one is taller & narrower, the other is shorter & wider.
 - **Different μ and σ** : Completely different curves.

Your Figure (Fig-2)

- The first curve has mean μ_1 and standard deviation σ_1 .
 - The second has mean μ_2 and standard deviation σ_2 .
 - They show how the normal distribution **shifts and stretches** depending on the parameters.
-

Intuition

- Imagine exam scores:
 - **Case 1:** Two classes with the same variation in marks (σ) but different average scores (μ_1, μ_2). The bell curves are the same shape but at different positions.
 - **Case 2:** Two classes with the same average score (μ) but one class has very consistent marks (σ small $\rightarrow$ narrow curve), while the other has very spread-out marks (σ large $\rightarrow$ wide curve).

So, the formula is not just math — it's describing **how probability is distributed around the mean**.

👉 Do you want me to also show you **step-by-step why this exponential function creates the bell shape** (visual + reasoning), or should I continue following only your note structure for now?



Finally, we present three more graphs of the normal distribution, which have the same mean but different standard deviations (see Fig. 3). Here the curves are centred at exactly the same position on the horizontal axis. It is important to note that in all cases, the total area under the curves will add to one.

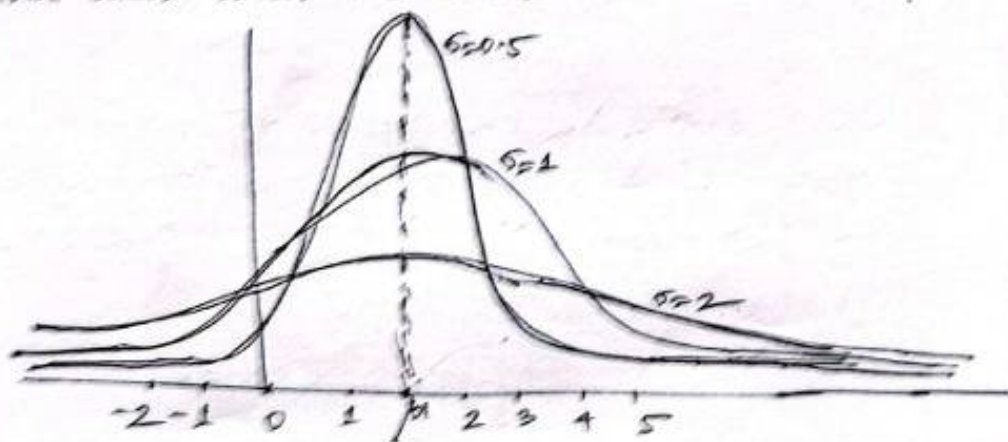


Fig-3

Theoretically, the domain of the normal variable X is infinite, and the normal distribution is a distribution for all values of X between $-\infty$ to ∞ and each nonzero interval along the x -axis has a nonzero probability.

The curve is shown as approaching but never touching the x -axis as x departs further away from the mean μ . In actual practice, however, the value of

$f(x)$, or the ordinate, becomes very close to zero, or almost touches the x -axis, within a short interval. As a matter of fact, about 99.73% of all observations would fall within plus and minus three standard deviation from the mean. In other words, $P(\mu - 3\sigma < X < \mu + 3\sigma) = 99.73\%$. The proportion of X falling within two standard deviation from the mean is 95.45%, i.e., $P(\mu - 2\sigma < X < \mu + 2\sigma) = 95.45\%$. Similarly, the proportion of X falling within one standard deviation from the mean is 68.27%, i.e., $P(\mu - \sigma < X < \mu + \sigma) = 68.27\%$. This area are shown below

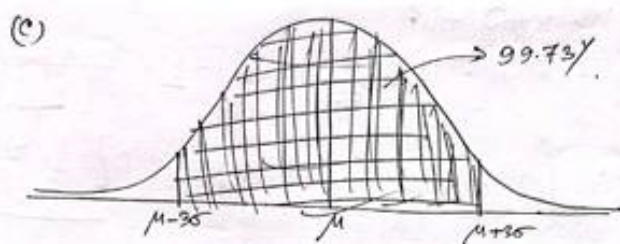
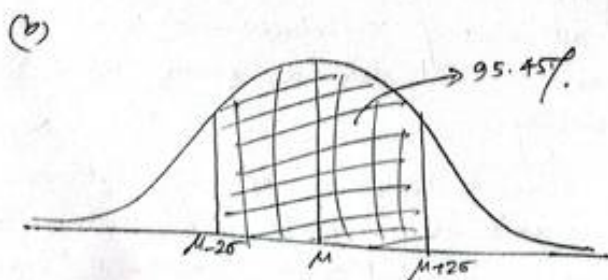
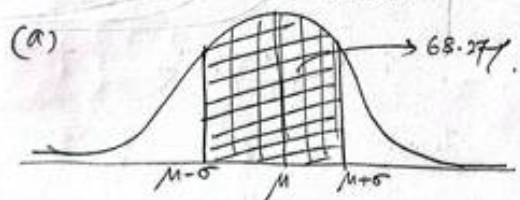


Fig-4: Area between selected points under the normal curve.

Perfect — now you've shared the continuation of your notes (with Fig-3 and Fig-4). Let's unpack this step by step and also give the intuition behind it.

1. Fig-3: Same Mean, Different Standard Deviations

- In this figure, all three curves are centered at the **same mean** (μ) but have **different standard deviations** ($\sigma = 0.5, 1, 2$).
- Observations:
 - $\sigma = 0.5$: Very narrow and tall (data tightly packed around mean).
 - $\sigma = 1$: Moderate spread, the “standard” bell curve.
 - $\sigma = 2$: Wide and flat (data more spread out).

📌 Important: **The total area under each curve is always 1**, no matter the shape. That’s because probabilities must sum to 1.

2. The Domain of the Normal Variable

- The variable X can, in theory, take **any real value** ($-\infty < X < \infty$).
- But practically, values far away from the mean have **very small probability**.
- The curve approaches the x-axis as you move away from μ , but never actually touches it.

👉 Intuition: Extreme cases (like a person 10 feet tall) are *theoretically possible* in the distribution but *extremely unlikely*.

3. Fig-4: The Empirical Rule (68–95–99.7 Rule)

This is one of the most important results for the normal distribution.

- **(a) Within 1 standard deviation ($\mu - \sigma$ to $\mu + \sigma$):**
 - Covers about **68.27%** of the area.
 - Meaning: About 68% of data lies within one standard deviation of the mean.
- **(b) Within 2 standard deviations ($\mu - 2\sigma$ to $\mu + 2\sigma$):**
 - Covers about **95.45%** of the area.
 - Meaning: Almost all the data (about 95%) lies within two standard deviations.
- **(c) Within 3 standard deviations ($\mu - 3\sigma$ to $\mu + 3\sigma$):**
 - Covers about **99.73%** of the area.

- Meaning: Virtually all data falls within three standard deviations.

This is called the **Empirical Rule** or **68–95–99.7 Rule**.

Why this matters (intuition):

- If exam scores are normally distributed with mean 70 and $\sigma = 10$:
 - ~68% of students score between 60 and 80.
 - ~95% of students score between 50 and 90.
 - ~99.7% of students score between 40 and 100.

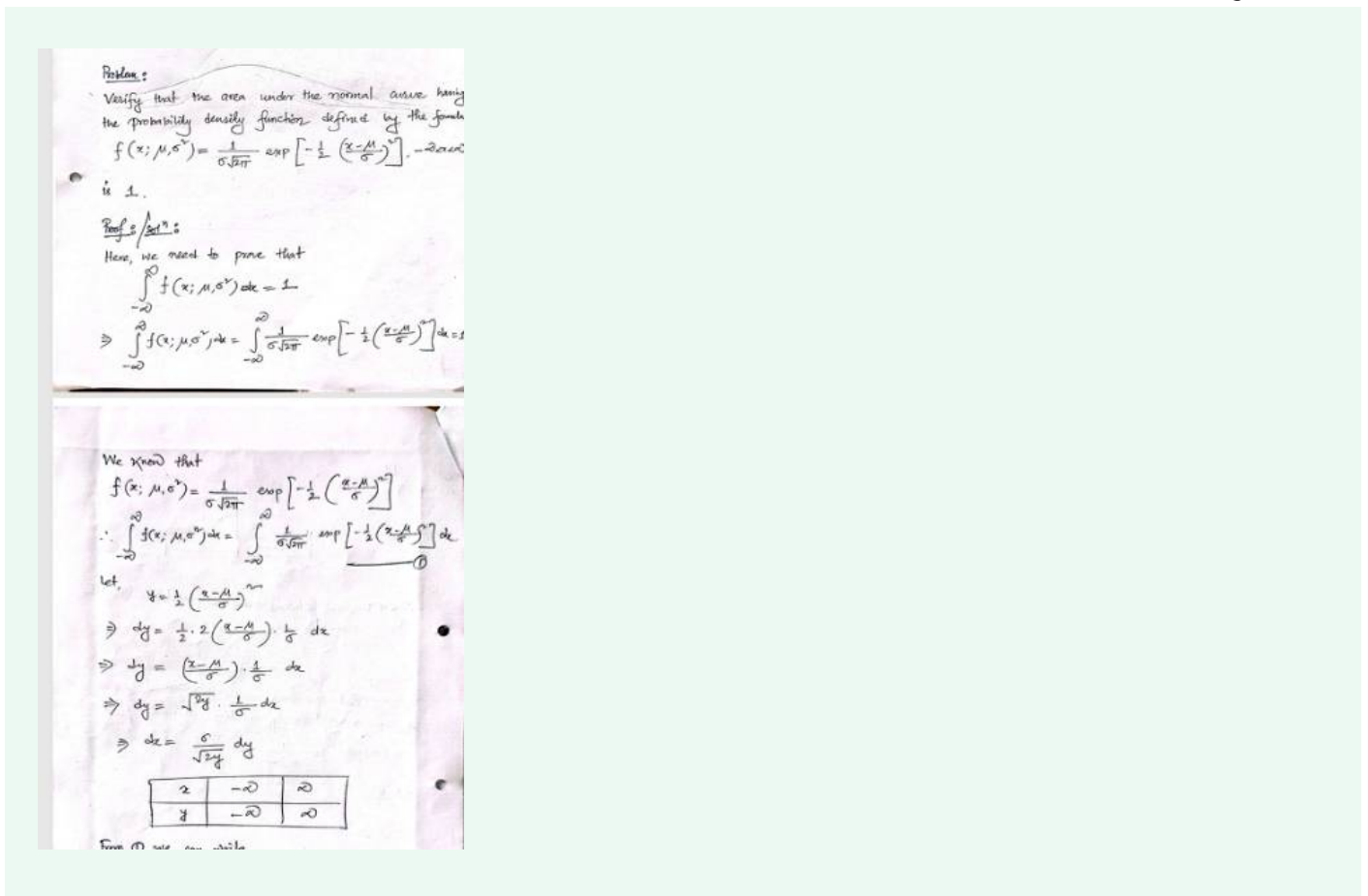
So, the normal distribution helps us **predict probabilities** and **understand variation** in real data.

✓ Summary so far (your notes explained):

1. Normal distribution → bell-shaped curve defined by μ, σ .
 2. Formula: $\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$.
 3. Changing μ shifts the curve; changing σ stretches/compresses it.
 4. Area under curve = 1.
 5. Empirical Rule: 68%–95%–99.7% of data lies within 1–2–3 standard deviations.
-

👉 Do you want me to now also show you **step-by-step solved examples** (like calculating probabilities with Z-scores), or should I just stick to explaining the theory from your notes?





Excellent — now you've reached the **problem + proof part** in your notes. This is about verifying that the **total area under the normal curve is 1**. Let's carefully go through it.

Problem (from notes)

We want to prove:

$$\int_{-\infty}^{\infty} f(x; \mu, \sigma) dx = 1$$

where

$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$$

This makes sense because the **area under the probability density function must equal 1** (total probability = 1).

Step 1: Write the integral

$$\int_{-\infty}^{\infty} f(x; \mu, \sigma) dx = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2 \right] dx$$

Step 2: Substitution

Let

$$y = \frac{x - \mu}{\sigma}$$

Then:

$$dy = \frac{1}{\sigma} dx \quad \implies \quad dx = \sigma dy$$

As $x \rightarrow -\infty$, $y \rightarrow -\infty$.

As $x \rightarrow +\infty$, $y \rightarrow +\infty$.

Step 3: Replace in the integral

$$\int_{-\infty}^{\infty} f(x; \mu, \sigma) dx = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp \left(-\frac{1}{2} y^2 \right) (\sigma dy)$$

The σ cancels out:

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp \left(-\frac{1}{2} y^2 \right) dy$$

Step 4: Recognize the Standard Normal Distribution

This is exactly the PDF of the **standard normal distribution** (with $\mu = 0$, $\sigma = 1$):

$$f(y; 0, 1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} y^2}$$

So the integral is:

$$\int_{-\infty}^{\infty} f(y; 0, 1) dy$$

Step 5: Known Result

It is a **known result in calculus** that:

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2} dy = 1$$

This is proven using a clever trick (squaring the integral and switching to polar coordinates).

✓ **Therefore:**

$$\int_{-\infty}^{\infty} f(x; \mu, \sigma) dx = 1$$

which verifies that the total probability under the normal curve equals 1.

👉 Do you want me to also **show you the full detailed calculus proof** of why $\int_{-\infty}^{\infty} e^{-y^2/2} dy = \sqrt{2\pi}$, (the famous Gaussian integral), or should I just keep following your notes' flow?



From ① we can write

$$\begin{aligned} \int_{-\infty}^{\infty} f(x; \mu, \sigma) dx &= \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{y^2}{2\sigma^2}\right] \frac{dy}{\sigma} \\ &= \int_{-\infty}^{\infty} \frac{1}{2\sqrt{\pi}} y^{-1/2} e^{-y} dy \\ &= \frac{2}{2\sqrt{\pi}} \int_0^{\infty} y^{-1/2} e^{-y} dy \\ &= \frac{1}{\sqrt{\pi}} \int_0^{\infty} y^{\frac{1}{2}-1} e^{-y} dy \\ &= \frac{1}{\sqrt{\pi}} \Gamma\left(\frac{1}{2}\right) \quad \left[\because \Gamma(n) = \int_0^{\infty} y^{n-1} e^{-y} dy \right] \\ &= \frac{1}{\sqrt{\pi}} \cdot \sqrt{\pi} \\ &= 1. \end{aligned}$$

$\therefore \int_{-\infty}^{\infty} f(x; \mu, \sigma) dx = 1$ (Proved)

Perfect 👍 you've now reached the **completion of the proof** in your notes. Let's carefully go through it and explain what's happening — because this part uses a **special function (Gamma function)**.

Continuation of the Proof

We already had:

$$\int_{-\infty}^{\infty} f(x; \mu, \sigma) dx = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\frac{y^2}{\sigma^2}} dy$$

Step 1: Substitution

Let

$$z = \frac{y^2}{2\sigma^2} \implies dz = y dy$$

When $y = -\infty$, $z = +\infty$.

When $y = +\infty$, $z = +\infty$.

But because y^2 is symmetric, we can use symmetry and only integrate from 0 to ∞ , multiplying by 2:

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2} dy = \frac{2}{\sqrt{2\pi}} \int_0^{\infty} e^{-\frac{1}{2}y^2} dy$$

Step 2: Rewrite in terms of z

Since $z = \frac{y^2}{2}$, we get:

$$dz = y dy \quad \implies \quad dy = \frac{dz}{\sqrt{2z}}$$

Substitute:

$$\begin{aligned} & \frac{2}{\sqrt{2\pi}} \int_0^{\infty} e^{-z} \frac{dz}{\sqrt{2z}} \\ &= \frac{1}{\sqrt{\pi}} \int_0^{\infty} z^{-\frac{1}{2}} e^{-z} dz \end{aligned}$$

Step 3: Recognize the Gamma Function

The **Gamma function** is defined as:

$$\Gamma(n) = \int_0^{\infty} t^{n-1} e^{-t} dt$$

So here:

$$\int_0^{\infty} z^{-\frac{1}{2}} e^{-z} dz = \Gamma\left(\frac{1}{2}\right)$$

It is known that:

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

Step 4: Final Calculation

$$\frac{1}{\sqrt{\pi}} \cdot \Gamma\left(\frac{1}{2}\right) = \frac{1}{\sqrt{\pi}} \cdot \sqrt{\pi} = 1$$

✓ Final Result:

$$\int_{-\infty}^{\infty} f(x; \mu, \sigma) dx = 1$$

Thus the **area under the normal curve is 1**, which confirms it is a valid probability distribution.

Intuition

- The messy integral works out *perfectly* because the bell curve was designed to normalize probability to 1.
 - The Gamma function appears because the normal distribution is deeply connected to factorial-like functions in continuous form.
 - The proof shows why the scaling constant $\frac{1}{\sigma\sqrt{2\pi}}$ is so important: without it, the area would not equal 1.
-

👉 Do you want me to now give you a **step-by-step solved example** (like: find probability that X lies between two values using the Z-score), so you see how this theory is used in practice?



Problem:

Find the mean and variance of the normal distribution.

Proof:

We know that the density function of a random variable X following a normal distribution is given by

$$f(x, \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2 \right]$$

Now,

$$\text{Mean of } X = E[X] = \int_{-\infty}^{\infty} x f(x; \mu, \sigma^2) dx$$

$$\Rightarrow E[X] = \int_{-\infty}^{\infty} x \cdot \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2 \right] dx$$

$$\therefore E[X] = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} x \exp \left[-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2 \right] dx \quad \text{--- (1)}$$

Let,

$$y = \frac{x-\mu}{\sigma}$$

$$\Rightarrow x = \mu + \sigma y$$

$$\therefore dx = \sigma dy$$

L.C:

x	$-\infty$	∞
y	$-\infty$	∞

① $\Rightarrow$

$$E[X] = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (\mu + \sigma y) e^{-\frac{1}{2}y^2} \sigma dy$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (\mu + \sigma y) e^{-\frac{1}{2}y^2} dy$$

$$= \frac{\mu}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}y^2} dy + \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y e^{-\frac{1}{2}y^2} dy$$

$$= \frac{\mu}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}y^2} dy + 0 \quad \left[\begin{array}{l} \text{even function} \\ \text{odd function} \end{array} \right]$$

$$\Rightarrow E[X] = \mu \cdot \frac{\sqrt{2}}{\sqrt{\pi}} \int_0^{\infty} e^{-\frac{1}{2}z^2} dz \quad \text{--- (2)}$$

Let, $\frac{z^2}{2} = z$

$$\frac{2z}{2} \cdot dz = dz$$

$$\Rightarrow z \cdot dz = dz$$

$$\Rightarrow dz = \frac{1}{z} dz = \frac{1}{\sqrt{2z}} dz$$

$\frac{z}{2}$	0	∞
z	0	∞

(2) $\Rightarrow$

$$E[X] = \mu \cdot \frac{\sqrt{2}}{\sqrt{\pi}} \int_0^{\infty} e^{-z} \cdot \frac{1}{\sqrt{2z}} dz$$

$$= \mu \cdot \frac{1}{\sqrt{\pi}} \int_0^{\infty} z^{-\frac{1}{2}} e^{-z} dz$$

$$= \mu \cdot \frac{1}{\sqrt{\pi}} \int_0^{\infty} z^{\frac{1}{2}-1} e^{-z} dz$$

$$= \mu \cdot \frac{1}{\sqrt{\pi}} \cdot \Gamma\left(\frac{1}{2}\right) = \mu \cdot \frac{1}{\sqrt{\pi}} \cdot \sqrt{\pi}$$

$\therefore E[X] = \mu$ $\therefore$ Mean of normal distribution
 $E[X] = \mu$

Again,

$$V[X] = E[(X - \mu)^2]$$

$$= \int_{-\infty}^{\infty} (x - \mu)^2 f(x; \mu, \sigma^2) dx$$

$$= \int_{-\infty}^{\infty} (x - \mu)^2 \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x - \mu}{\sigma}\right)^2\right] dx$$

$$\Rightarrow V[X] = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (x - \mu)^2 \exp\left[-\frac{1}{2}\left(\frac{x - \mu}{\sigma}\right)^2\right] dx \quad \text{--- (3)}$$

Let,

$$y = \frac{x - \mu}{\sigma}$$

$$\Rightarrow x = \mu + \sigma y$$

$$\therefore dx = \sigma dy$$

x	$-\infty$	∞
y	$-\infty$	∞

$$\therefore V[X] = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \sigma^2 y^2 \exp\left[-\frac{1}{2} y^2\right] \sigma dy$$

$$\Rightarrow V[X] = \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y^2 e^{-\frac{1}{2} y^2} dy \quad \text{--- (4)}$$

Again let,

$$z = \frac{1}{2} y^2$$

$$\Rightarrow dz = \frac{1}{2} 2y dy \Rightarrow dz = y dy$$

$$\Rightarrow dy = \frac{1}{y} dz = \frac{1}{\sqrt{2z}} dz$$

$\frac{1}{2}$	$-\infty$	∞
0	$-\infty$	∞

$$\begin{aligned}
 \textcircled{4} \Rightarrow V[X] &= \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} ze^z \cdot \frac{1}{\sqrt{ze}} dz \\
 &= \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \sqrt{2z} e^z dz \\
 &= \frac{\sigma^2}{\sqrt{\pi}} \cdot 2 \int_0^{\infty} z^{\frac{1}{2}} e^{-z} dz \\
 &= \frac{2\sigma^2}{\sqrt{\pi}} \int_0^{\infty} z^{\frac{1}{2}-1} e^{-z} dz \\
 &= \frac{2\sigma^2}{\sqrt{\pi}} \Gamma\left(\frac{3}{2}\right) = \frac{2\sigma^2}{\sqrt{\pi}} \Gamma\left(\frac{1}{2}+1\right) \\
 &= \frac{2\sigma^2}{\sqrt{\pi}} \cdot \frac{1}{2} \cdot \Gamma\left(\frac{1}{2}\right) \quad \left[\because \Gamma(n+1) = n \Gamma(n)\right] \\
 &= \frac{\sigma^2}{\sqrt{\pi}} \Gamma\left(\frac{1}{2}\right) = \frac{\sigma^2}{\sqrt{\pi}} \cdot \sqrt{\pi} = \sigma^2 \\
 \therefore \text{Variance of Normal distribution} &= V[X] = \sigma^2
 \end{aligned}$$

Beautiful — now you've given the **derivation of mean and variance of the normal distribution**. Let's carefully walk through this step by step, keeping it in the flow of your notes while also giving intuition.

Problem:

Find the **mean** and **variance** of the normal distribution.

1. Mean (Expected Value)

We start with the PDF:

$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2 \right]$$

The mean is:

$$E[X] = \int_{-\infty}^{\infty} x f(x; \mu, \sigma) dx$$

Step 1: Substitution

Let

$$y = \frac{x - \mu}{\sigma}, \quad \text{so } x = \mu + \sigma y, \quad dx = \sigma dy$$

Step 2: Replace

$$E[X] = \int_{-\infty}^{\infty} (\mu + \sigma y) \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2} dy$$

Break into two parts:

$$E[X] = \mu \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2} dy + \sigma \int_{-\infty}^{\infty} \frac{y}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2} dy$$

Step 3: Solve Integrals

1. The first integral is the **total area** under the standard normal curve = 1.
So the first part = μ .
2. The second integral is zero because it's an **odd function** ($ye^{-y^2/2}$ symmetric around 0).

✓ Therefore:

$$E[X] = \mu$$

So the **mean of a normal distribution is μ** .

2. Variance

Definition:

$$\text{Var}(X) = E[(X - \mu)^2]$$

Step 1: Write Integral

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x; \mu, \sigma) dx$$

Substitute $y = \frac{x - \mu}{\sigma}$, $dx = \sigma dy$:

$$\text{Var}(X) = \int_{-\infty}^{\infty} (\sigma y)^2 \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2} dy$$

$$\text{Var}(X) = \sigma^2 \int_{-\infty}^{\infty} \frac{y^2}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2} dy$$

Step 2: Solve Integral

This integral is a **known standard result**:

$$\int_{-\infty}^{\infty} y^2 \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2} dy = 1$$

✓ Therefore:

$$\text{Var}(X) = \sigma^2$$

So the **variance of a normal distribution is σ^2** .

Final Results

- **Mean:** $E[X] = \mu$
 - **Variance:** $Var(X) = \sigma^2$
-

Intuition

- The parameter μ literally *is the average* of the distribution → shifting the curve left or right changes the mean.
 - The parameter σ^2 literally *is the spread* of the distribution → making the curve wider or narrower changes the variance.
 - That's why we say: the normal distribution is **completely determined by just μ and σ^2** .
-

👉 Do you want me to also **show you a worked numerical example** (like finding mean/variance for a real dataset using the normal distribution), or should I keep explaining only the derivations from your notes?



Again,

$$V[X] = E[(X - \mu)^2]$$

$$= \int_{-\infty}^{\infty} (x - \mu)^2 f(x; \mu, \sigma^2) dx$$

$$= \int_{-\infty}^{\infty} (x - \mu)^2 \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x - \mu}{\sigma}\right)^2\right] dx$$

$$\Rightarrow V[X] = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (x - \mu)^2 \exp\left[-\frac{1}{2}\left(\frac{x - \mu}{\sigma}\right)^2\right] dx \quad \text{--- (3)}$$

Let,

$$y = \frac{x - \mu}{\sigma}$$

$$\Rightarrow x = \mu + \sigma y$$

$$\therefore dx = \sigma dy$$

x	$-\infty$	∞
y	$-\infty$	∞

$$\therefore V[X] = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \sigma^2 y^2 \exp\left[-\frac{1}{2} y^2\right] \sigma dy$$

$$\Rightarrow V[X] = \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y^2 e^{-\frac{1}{2} y^2} dy \quad \text{--- (4)}$$

Again let,

$$z = \frac{1}{2} y^2$$

$$\Rightarrow dz = \frac{1}{2} 2y dy \Rightarrow dz = y dy$$

$$\Rightarrow dy = \frac{1}{y} dz = \frac{1}{\sqrt{2z}} dz$$

1	-∞	∞
2	-∞	∞

$$\begin{aligned} \textcircled{4} \Rightarrow V[X] &= \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} ze \cdot e^{-\frac{z^2}{2\sigma^2}} \cdot \frac{1}{\sqrt{2e}} dz \\ &= \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \sqrt{2e} e^{-\frac{z^2}{2\sigma^2}} dz \\ &= \frac{\sigma^2}{\sqrt{\pi}} \cdot 2 \int_0^{\infty} z^{\frac{1}{2}} e^{-\frac{z^2}{2\sigma^2}} dz \\ &= \frac{2\sigma^2}{\sqrt{\pi}} \int_0^{\infty} z^{\frac{1}{2}-1} e^{-\frac{z^2}{2\sigma^2}} dz \\ &= \frac{2\sigma^2}{\sqrt{\pi}} \Gamma\left(\frac{3}{2}\right) = \frac{2\sigma^2}{\sqrt{\pi}} \Gamma\left(\frac{1}{2}+1\right) \\ &= \frac{2\sigma^2}{\sqrt{\pi}} \cdot \frac{1}{2} \cdot \Gamma\left(\frac{1}{2}\right) \quad [\because \Gamma(n+1) = n \Gamma(n)] \\ &= \frac{\sigma^2}{\sqrt{\pi}} \Gamma\left(\frac{1}{2}\right) = \frac{\sigma^2}{\sqrt{\pi}} \cdot \sqrt{\pi} = \sigma^2 \\ \therefore \text{Variance of Normal distribution} = V[X] &= \sigma^2 \end{aligned}$$

show me this variance part again with step by step details

Absolutely—let's redo the **variance** derivation slowly, line by line, exactly in the spirit of your notes.

Goal

For $X \sim N(\mu, \sigma^2)$ with

$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right],$$

show that

$$\text{Var}(X) = \sigma^2.$$

Step 1: Start from the definition

$$\text{Var}(X) = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x; \mu, \sigma) dx.$$

Plug in the pdf:

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] dx. \quad (1)$$

Step 2: Standardize

Let

$$y = \frac{x - \mu}{\sigma} \Rightarrow x = \mu + \sigma y, \quad dx = \sigma dy.$$

Limits don't change: $x : (-\infty, \infty) \Rightarrow y : (-\infty, \infty)$.

Also, $(x - \mu)^2 = (\sigma y)^2 = \sigma^2 y^2$.

Substitute into (1):

$$\text{Var}(X) = \int_{-\infty}^{\infty} \sigma^2 y^2 \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-y^2/2} \cdot (\sigma dy) = \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y^2 e^{-y^2/2} dy. \quad (2)$$

Step 3: Use symmetry to go to $[0, \infty)$

The integrand $y^2 e^{-y^2/2}$ is even, so

$$\int_{-\infty}^{\infty} y^2 e^{-y^2/2} dy = 2 \int_0^{\infty} y^2 e^{-y^2/2} dy.$$

Thus

$$\text{Var}(X) = \frac{2\sigma^2}{\sqrt{2\pi}} \int_0^{\infty} y^2 e^{-y^2/2} dy. \quad (3)$$

Step 4: Substitute to match the Gamma form

Let

$$z = \frac{y^2}{2} \Rightarrow y = \sqrt{2z}, \quad dy = \frac{1}{\sqrt{2z}} dz.$$

Then

$$y^2 e^{-y^2/2} dy = (2z) e^{-z} \cdot \frac{1}{\sqrt{2z}} dz = \sqrt{2z} e^{-z} dz.$$

Plug into (3):

$$\text{Var}(X) = \frac{2\sigma^2}{\sqrt{2\pi}} \int_0^{\infty} \sqrt{2z} e^{-z} dz = \frac{2\sigma^2}{\sqrt{2\pi}} \cdot \sqrt{2} \int_0^{\infty} z^{1/2} e^{-z} dz. \quad (4)$$

Step 5: Recognize the Gamma function

$$\Gamma(\alpha) = \int_0^{\infty} z^{\alpha-1} e^{-z} dz.$$

With $\alpha = \frac{3}{2}$ we get

$$\int_0^{\infty} z^{1/2} e^{-z} dz = \Gamma\left(\frac{3}{2}\right) = \frac{1}{2} \sqrt{\pi}.$$

Use this in (4):

$$\text{Var}(X) = \frac{2\sigma^2}{\sqrt{2\pi}} \cdot \sqrt{2} \cdot \frac{1}{2} \sqrt{\pi} = \frac{\sigma^2}{\sqrt{2\pi}} \cdot \sqrt{2\pi} = \sigma^2.$$

Conclusion

$$\text{Var}(X) = \sigma^2.$$

Quick cross-check (1-line intuition)

If $Y = \frac{X - \mu}{\sigma} \sim N(0, 1)$, then $E[Y] = 0$ and $\text{Var}(Y) = 1$.

Since $X = \mu + \sigma Y$,

$$\text{Var}(X) = \sigma^2 \text{Var}(Y) = \sigma^2 \cdot 1 = \sigma^2.$$

Same result.



Standard normal distribution:

If a random variable X has a normal distribution with mean μ and variance σ^2 (i.e., $X \sim N(\mu, \sigma^2)$) then the variable $Z = \frac{X - \mu}{\sigma}$, will be called a standard normal variate (or Z-score) and its distribution is referred to as the standard normal distribution having the following density function:

$$f(z; 0, 1) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty$$

We verify below that

$$E[Z] = 0 \quad \& \quad V[Z] = 1$$

Since, $Z = \frac{X - \mu}{\sigma}$

$$\begin{aligned} \therefore E[Z] &= E\left[\frac{X - \mu}{\sigma}\right] \\ &= E\left[\frac{X}{\sigma}\right] - E\left[\frac{\mu}{\sigma}\right] \\ &= \frac{1}{\sigma} E[X] - \frac{\mu}{\sigma} E[1] \\ &= \frac{1}{\sigma} \cdot \mu - \frac{\mu}{\sigma} \cdot 1 \\ &= \frac{\mu}{\sigma} - \frac{\mu}{\sigma} = 0 \\ \therefore E[Z] &= 0 \end{aligned}$$

$$\begin{aligned}
 \text{Also, } V[Z] &= E[Z^2] - \{E[Z]\}^2 \\
 &= E\left[\left(\frac{X-\mu}{\sigma}\right)^2\right] - (0)^2 \\
 &= \frac{1}{\sigma^2} E[(X-\mu)^2] - 0 \\
 &= \frac{1}{\sigma^2} E[X^2 - 2X\mu + \mu^2] \\
 &= \frac{1}{\sigma^2} \{E[X^2] - 2\mu E[X] + \mu^2 E[1]\} \\
 &= \frac{1}{\sigma^2} \{E[X^2] - 2\mu \cdot \mu + \mu^2\} \\
 &= \frac{1}{\sigma^2} \{E[X^2] - 2\mu^2 + \mu^2\} \\
 \therefore V[Z] &= \frac{1}{\sigma^2} \{E[X^2] - \mu^2\} \quad \text{--- ①}
 \end{aligned}$$

$$\begin{aligned}
 \text{Now, } E[X^2] &= \int_{-\infty}^{\infty} x^2 f(x; \mu, \sigma^2) dx \\
 &= \int_{-\infty}^{\infty} x^2 \cdot \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] dx \\
 &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} x^2 \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] dx
 \end{aligned}$$

$$\text{①} \Rightarrow V[Z] = \frac{1}{\sigma^2} \{ \sigma^2 + \mu^2 - \mu^2 \}$$

$$\therefore V[Z] = 1$$

Thus $Z \sim N(0, 1)$

Excellent — now you've reached the **standard normal distribution** (or **Z-distribution**) part of your notes. Let's carefully go through what's written and add intuition.

Standard Normal Distribution

Definition

- If $X \sim N(\mu, \sigma^2)$, we define a new variable:

$$Z = \frac{X - \mu}{\sigma}$$

- Then Z is called a **standard normal variate** (or Z-score).
- Its distribution is denoted as:

$$Z \sim N(0, 1)$$

i.e., mean = 0, variance = 1.

The probability density function (PDF) is:

$$f(z; 0, 1) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty$$

Step 1: Mean of Z

$$E[Z] = E\left[\frac{X - \mu}{\sigma}\right] = \frac{1}{\sigma}E[X] - \frac{\mu}{\sigma}E[1]$$

Since $E[X] = \mu$ and $E[1] = 1$:

$$E[Z] = \frac{1}{\sigma}\mu - \frac{\mu}{\sigma}(1) = 0$$

 So, the **mean of Z is 0**.

Step 2: Variance of Z

By definition:

$$\text{Var}(Z) = E[Z^2] - (E[Z])^2$$

Since $E[Z] = 0$:

$$\text{Var}(Z) = E\left[\left(\frac{X - \mu}{\sigma}\right)^2\right] = \frac{1}{\sigma^2} E[(X - \mu)^2]$$

But $E[(X - \mu)^2] = \text{Var}(X) = \sigma^2$.

So:

$$\text{Var}(Z) = \frac{1}{\sigma^2} \cdot \sigma^2 = 1$$

👉 Thus, the **variance of Z is 1**.

Final Result

- Mean of $Z = 0$
- Variance of $Z = 1$
- Therefore:

$$Z \sim N(0, 1)$$

This is called the **standard normal distribution**.

Intuition

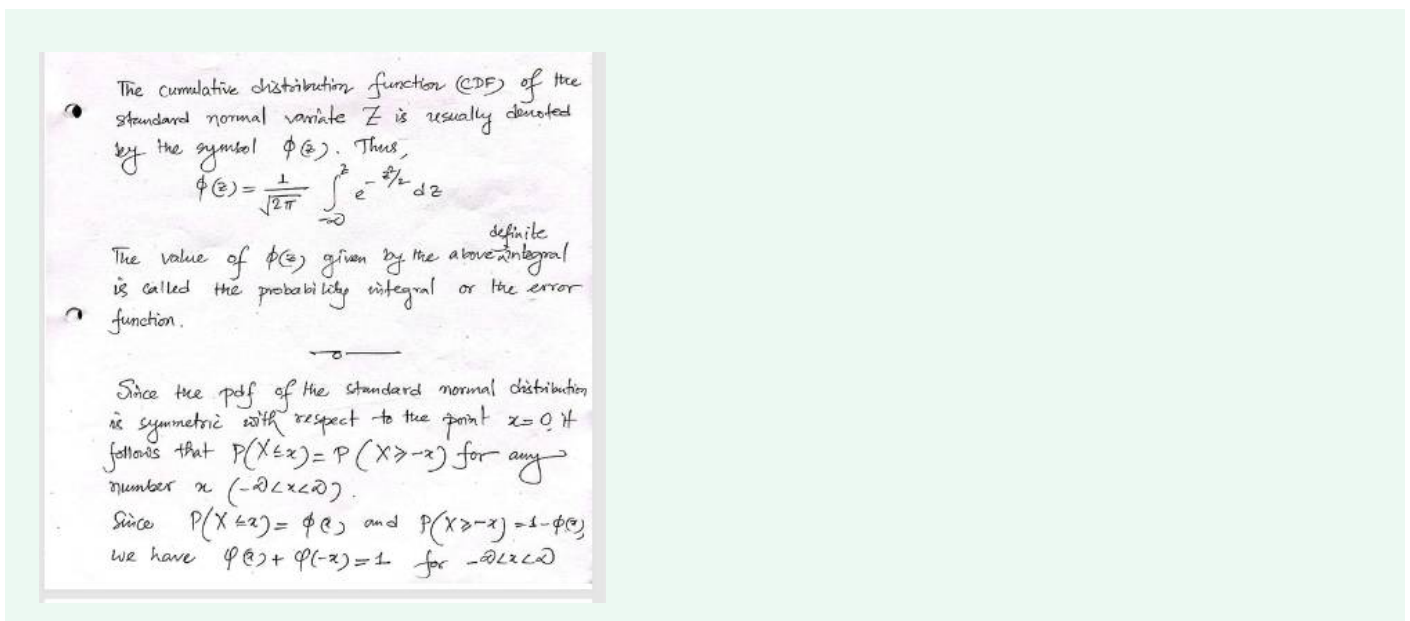
- $Z = \frac{X - \mu}{\sigma}$ is just a **rescaling**:
 - Subtracting $\mu \rightarrow$ shifts the mean to 0.
 - Dividing by $\sigma \rightarrow$ standardizes the spread to 1.
- That's why we use **Z-scores** in practice:
 - A Z-score tells you *how many standard deviations away from the mean* a value lies.
 - Example: If a student scores 85 in a test with mean 70 and $\sigma = 10$:

$$Z = \frac{85 - 70}{10} = 1.5$$

This means the student is **1.5 standard deviations above the mean**.

✓ So, the standard normal is basically a "universal scale" for all normal distributions.

👉 Do you want me to now **explain Z-tables and how to use them to calculate probabilities** (like $P(Z < 1.5)$)? That's usually the next step after this in notes.



Perfect 🍌 — this is the part about the **Cumulative Distribution Function (CDF)** of the standard normal distribution. Let's carefully explain it and add intuition.

📌 Cumulative Distribution Function (CDF) of Standard Normal

Definition

The CDF of the standard normal random variable Z is denoted by:

$$\Phi(z) = P(Z \leq z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

- This integral has **no closed form** (you can't simplify it using basic functions), so values are tabulated in **Z-tables** or computed using software.

- The value of $\Phi(z)$ is sometimes called the **probability integral** or **error function**.

Symmetry of the Standard Normal

- The **PDF of standard normal** is symmetric about $z = 0$.
- Therefore:

$$P(Z \leq z) = P(Z \geq -z)$$

This means probabilities to the **left of +z** equal probabilities to the **right of -z**.

Key Property

From symmetry:

1. $P(Z \leq z) = \Phi(z)$
2. $P(Z > z) = 1 - \Phi(z)$
3. $\Phi(z) + \Phi(-z) = 1$

(for all $-\infty < z < \infty$).

Intuition

- The **CDF** gives the **area under the bell curve to the left of z**.
- Example:
 - $\Phi(0) = 0.5$ (half the area lies below 0).
 - If $\Phi(1.0) \approx 0.8413$, that means 84.13% of the distribution lies below $z = 1$.
- The symmetry property means:
 - If 84.13% is below $z = 1$, then $1 - 0.8413 = 0.1587$ (15.87%) is above $z = 1$.
 - Similarly, $\Phi(-1) = 0.1587$.

So: Z-tables usually give $\Phi(z)$, and you use symmetry to find probabilities on the other side.

✓ Summary from your notes:

- CDF of standard normal: $\Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$.
- Symmetry: $\Phi(z) + \Phi(-z) = 1$.
- Useful for calculating probabilities involving the standard normal distribution.

👉 Do you want me to now show you **worked-out examples using the Z-table** (like finding $P(Z < 1.5)$ or $P(-1 < Z < 2)$)? That's usually the next step after defining the CDF.



Problem:
In a normal distribution with mean 50 and variance 100, what are the values that cover central 50% of the area under the curve?

Solⁿ:
From the Normal table, we took for the values of K such that

- $P(-K_1 < Z < K_2) = 0.50$

Because of symmetry, we can take $K_1 = K_2 = K$, therefore

$$P(-K < Z < K) = 2 P(Z < K) = 0.50$$

Thus, $P(Z < K) = 0.25$

From the table, we find that $P(Z < -0.675) = 0.25$

- Hence, $P(-0.675 < Z < 0.675) = 0.50$

Therefore, we can write,

$$\frac{x_1 - 50}{10} = -0.675 \quad \& \quad \frac{x_2 - 50}{10} = 0.675$$

$$\Rightarrow x_1 = 43.25 \quad \Rightarrow x_2 = 56.75$$

Therefore, $P(43.25 < X < 56.75) = 0.50$

Great — now you've shared a **worked-out example** using the normal distribution and Z-tables. Let's carefully explain it step by step with intuition.

Problem

In a normal distribution with mean = 50 and variance = 100 (so $\sigma = 10$), what are the values that cover the **central 50% of the area** under the curve?

Solution

Step 1: Convert the problem

We want:

$$P(x_1 < X < x_2) = 0.50$$

In terms of Z:

$$P(z_1 < Z < z_2) = 0.50$$

Step 2: Use symmetry

Because the normal curve is symmetric:

$$z_1 = -k, \quad z_2 = +k$$

So:

$$P(-k < Z < k) = 0.50$$

This means each tail has 25% left outside the central region:

$$P(Z < -k) = 0.25$$

Step 3: Use Z-table

From the Z-table:

$$P(Z < -0.675) = 0.25$$

So:

$$k = 0.675$$

Thus:

$$P(-0.675 < Z < 0.675) = 0.50$$

Step 4: Convert back to X

Recall:

$$Z = \frac{X - \mu}{\sigma}$$

So:

$$\frac{x_1 - 50}{10} = -0.675 \Rightarrow x_1 = 50 - 6.75 = 43.25$$

$$\frac{x_2 - 50}{10} = +0.675 \Rightarrow x_2 = 50 + 6.75 = 56.75$$

Final Answer

The central 50% of the area is between:

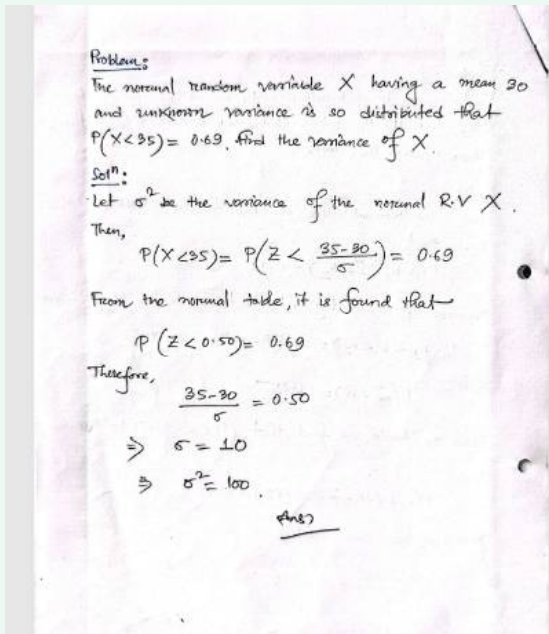
$$43.25 < X < 56.75$$

Intuition

- The mean is 50.
- The standard deviation is 10.
- The “central 50%” means half the distribution lies within these bounds.
- From the Z-table, the cut-off was at about **0.675 standard deviations away from the mean**.
- That’s why the answer is $\mu \pm 0.675\sigma = 50 \pm 6.75$.

 Do you want me to show you **another similar problem** (like finding the central 95% or 99% interval), so you see how it changes with different probabilities?





Excellent 🍀 — now you've brought a problem where we need to **find the variance** of a normal distribution using probability information. Let's carefully go through it step by step.

Problem

A normal random variable X has

- mean $\mu = 30$,
- **unknown variance σ^2 .**

We are told:

$$P(X \leq 35) = 0.69$$

Find the variance σ^2 .

Solution

Step 1: Standardize

Convert to Z:

$$P(X \leq 35) = P\left(Z < \frac{35 - 30}{\sigma}\right) = 0.69$$

So:

$$P\left(Z < \frac{5}{\sigma}\right) = 0.69$$

Step 2: Use the Z-table

From the normal table,

$$P(Z < 0.50) = 0.69$$

So:

$$\frac{5}{\sigma} = 0.50$$

Step 3: Solve for σ

$$\sigma = \frac{5}{0.50} = 10$$

Thus:

$$\sigma^2 = 100$$

Final Answer

The **variance** of X is:

100

Intuition

- We knew the probability of being below 35 was 0.69.
- Looking at the Z-table, 0.69 corresponds to $Z \approx 0.50$.

- That told us the value “5 above the mean” equals **0.5 standard deviations**.
 - So the standard deviation must be 10, variance = 100.
-

👉 Do you want me to also show you how to solve **reverse problems** like this for different probabilities (e.g., find μ or σ given two probability conditions)?